

Stacks

Start Assignment

- Due Sunday by 11:59pm
- Points 10
- Submitting a text entry box or a file upload
- Available Sep 13 at 12am - Sep 23 at 11:59pm

(Thanks to Dr. David Chays of Adelphi University for his version of this assignment.)

Download the handout.

Write a program named `TestStack.java` that implements the following method to merge two stacks into a single stack.

```
/**
 * merge: Given two stacks containing Integer objects in increasing order from the bottom up,
 * create a third stack such that the Integer objects are in increasing order from the bottom up.
 * If an item appears n times in the two given stacks, it will appear n times in the new stack.
 *
 * @param s1 the first stack
 * @param s2 the second stack
 * @return the new stack, with the items from the two given stacks merged.
 */
```

The program will also implement the `reverse()` method:

```
/**
 * reverse: Given a stack of Integer objects, create and return a stack with the
 * same objects in reverse order.
 *
 * @param s1 the input stack
 * @return the new stack, with the items in reverse order
 */
```

Implement the `makeStack()` method to create a stack conveniently:

```
/**
 * makeStack: Given an array of Integer objects, create and return a stack with
 * the same objects. The first item in the array will be at the bottom of the
 * stack, the last item at the top.
 *
 * @param arr the Integer array
 * @return the new stack
 */
```

Your `main()` method must test the `merge()` method on at least these cases:

- Two stacks of equal length and no duplicate values.
- Two stacks of unequal length and no duplicate values.
- Two stacks of unequal length with duplicate values shared between stacks (for example `3 5 7 19` and `1 7 11 11 15`)

- One stack with values and one empty stack.

You will use the `Stack.java` code from the handout.

Note

The `merge()` and `reverse()` methods are not allowed to modify the input stack(s). You may want to write the code so that the input stack(s) remain unchanged. You can use the `clone()` instance method in [Stack.java \(https://sjeccd.instructure.com/courses/53429/files/12442762?wrap=1\)](https://sjeccd.instructure.com/courses/53429/files/12442762?wrap=1)  [\(https://sjeccd.instructure.com/courses/53429/files/12442762/download?download_frd=1\)](https://sjeccd.instructure.com/courses/53429/files/12442762/download?download_frd=1) to make a duplicate copy of a stack.

When You Finish

Submit the `TestStack.java` file.

Stacks			
Criteria	Ratings		Pts
Readability Javadoc Comments + General structure of the code	2 pts Good	0 pts Poor	2 pts
Correctness All tests pass	8 pts Good	0 pts Poor	8 pts
			Total Points: 10